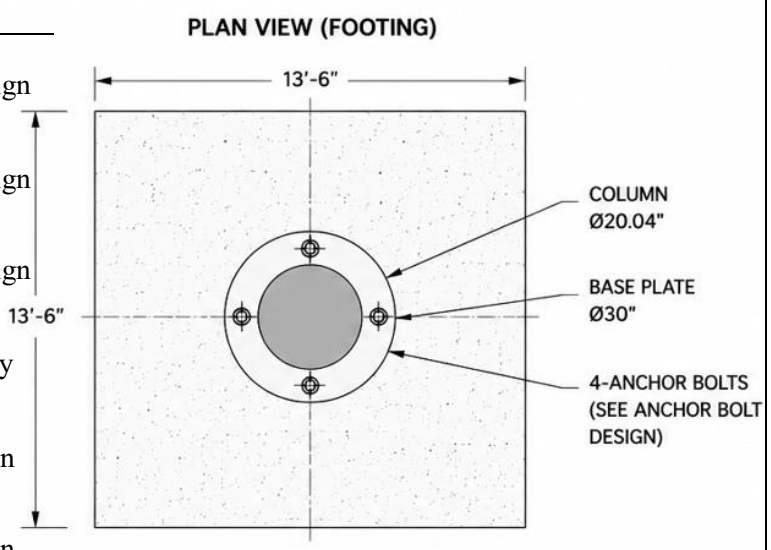
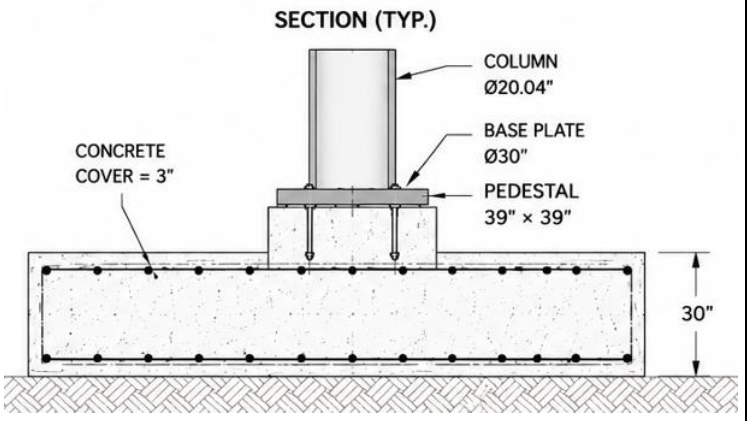
4. Footing segment

No	Design Parameter	Symbol	Value	Source
1	Total factored axial load	P_u	1637.5732kips	Main Tank Design
2	Factored overturning moment	M_u	5673.437kips-ft	Main Tank Design
3	Factored shear	V_u	57.218 kips	Main Tank Design
4	Number of legs	N_{leg}	4	Tank Geometry
5	Pedestal size	$B_p \times L_p$	39 × 39 in	Pedestal Design
6	Pedestal height	H_p	5'-6" = 66 in	Pedestal Design
7	Concrete strength	f'_c	4 ksi	User Input
8	Reinforcing steel	f_y	60 ksi	User Input
9	Concrete unit weight	γ_c	150 pcf	From ACI Code
10	Assumed footing thickness	h	30 in	Assumed
11	Concrete cover	c_c	3 in	Assumed
12	Bottom reinforcement		#8 bars, each way	User input
13	Allowable soil pressure (Assumed)	q_{allow}	3.0 ksf	User Input





Step 1: Footing Load

Gravity load per leg

$$P_{\text{leg}} = \frac{P_u}{N_{\text{leg}}}$$

$$P_{leg} = \frac{1637.5732}{4} = 409.3933 \text{ kip}$$

Self-weight of Pedestal

$$W_p = \gamma_c \left(b_p * h_p * L_p\right) = (0.150)(3.25)(3.25)(5.50)$$
$$W_p = 8.7141 \text{ kip}$$

Factored self-weight, assuming a 1.2 dead-load factor:

$$W_{p,u} = 1.2(8.7141) = 10.4569 \text{ kips}$$

$$P_c = P_{leg} + W_p$$
$$P_c = 409.3933 + 10.4569 = 419.8502 \text{ kip}$$

$$\boldsymbol{P_u = 419.8502 \text{ kip}}$$

This value is used consistently throughout the footing strength calculations.

Step 2: Preliminary Footing Size

Required footing area

$$\text{A} = \frac{P}{q_{allow}}$$

$$A_{req} = \frac{419.8502}{3} = 139.95 ft^2$$

$$B_{\text{req}} = \sqrt{139.95} = 11.83 \text{ ft} = 12\text{ft}$$

$$\boldsymbol{B = L = 12.0 \text{ ft}}$$

Footing Area:

$$A_f = 12(12) = 144 \text{ ft}^2$$

For considering in various condition, the size of the footing has been considered based on the drawing area and checked it:

$$B = L = 13.5 \text{ ft}$$

$$A_f = 13.5(13.5)$$

$$\boldsymbol{A_f = 182.25 \text{ ft}^2}$$

Note: This sizing uses a factored load with an allowable soil pressure, which is not a consistent final design check. Final sizing must use the applicable service-load combination for allowable bearing and include the overturning-moment effect.

Step 3: Soil Bearing Check

Average factored bearing pressure:

$$q_u = \frac{P_u}{A_f}$$

$$\text{q} = \frac{419.8502}{12 \text{ x } 13.5}$$

$$\text{q} = 2.303 \text{ ksf}$$

$$\text{q}_{\text{allow}} = 3.0 \text{ ksf}$$

$2.303\text{ksf} < 3.0 \text{ ksf}$…….. Bearing Pressure Pass

Step 4: Footing Thickness

Adopt:

$$h_{trail} = 30 \text{ in.}$$

Note: The software shall subsequently perform the required structural checks using the selected footing thickness. If any required check does not satisfy the applicable ACI 318 provisions, the software shall display a **warning/red alert** and the user shall increase the footing thickness $h$and repeat the checks until all required criteria are satisfied. (Recommended 25% - 30%h_{trail}).

Given:

$$\text{B} = \text{L} = 13.5 \text{ ft}$$

$$\text{c} = 39 \text{ in.}$$

$$\text{f}'_\text{c} = 4.0 \text{ ksi}$$

Bottom reinforcement (User Input) $\rho_{b_1} = \rho_{b_2}$:

#8 @ s each way

Concrete cover: Concrete cover shall be determined in accordance with the applicable

Concrete exposure	Member	Reinforcement	Specified cover, mm
Cast against and permanently in contact with ground	All	All	75
Exposed to weather or in contact with ground	All	No. 19 through No. 57 bars	50
		No. 16 bar, MW200 or MD200 wire, and smaller	40
Not exposed to weather or in contact with ground	Slabs, joists and walls	No. 43 and No. 57 bars	40
		No. 36 bar and smaller	20
	Beams, columns, pedestals and tension ties	Primary reinforcement, stirrups, ties, spirals and hoops	40

Table 20.6.1.3.1 Specified concrete cover for cast-in-place nonprestressed concrete members

ACI 318 requirements table 20.6.1.3.1 for the footing exposure and construction condition.

$$3 \text{ in.}$$

For #8 bars use Table A-1:

$d_b = 1.0$ in.

Therefore:

$$d = h - \text{cover} - \frac{d_b}{2}$$

$$d = 30 - 3 - \frac{1.0}{2}$$

$$d = 26.5 \text{ in.}$$

Step 5: Two-Way Shear / Punching Shear

[ACI 318-19: 22.6.4.1, 22.6.5.2 & 3]

For a square pedestal, the critical punching-shear perimeter is located approximately $\frac{d}{2}$ from the pedestal face.

$$d = 26.5 \text{ in.}$$

$$b_o = 4(c + d)$$

$$b_o = 4(39 + 26.5)$$

$$b_o = 262 \text{ in} = 21.833 \text{ ft}$$

or:

$$b_o = \frac{262}{12} = 21.83 \text{ ft}$$

$$V_u = P_u - q_u(c + d)^2$$

$$P_u = 419.8502 \text{ kip}$$

$$q_u = \frac{p_u}{A_f}$$

$$P_u = 419.8502 \text{ kip}$$

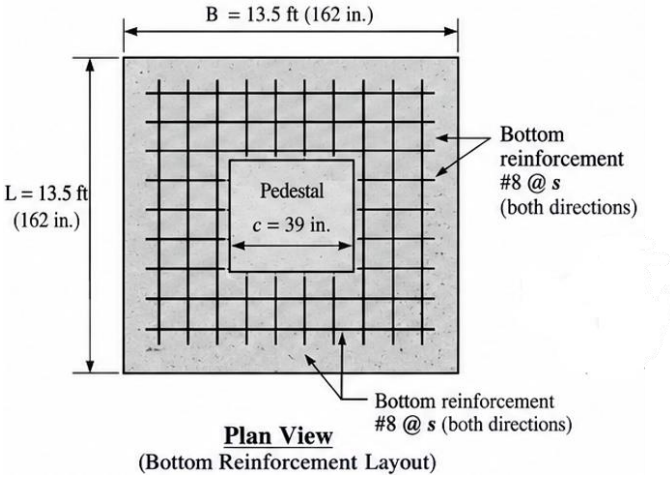
$$q_u = \frac{419.8502}{182.25} = 2.303$$

$$q_u = 2.303 \text{ ksf}$$

ASTM Standard Reinforcing Bar — Nominal Diameter

Bar designation	Nominal diameter, d_b (in.)
#3	0.375
#4	0.500
#5	0.625
#6	0.750
#7	0.875
#8	**1.000**
#9	1.128
#10	1.270
#11	1.410
#14	1.693
#18	2.257

Source: ACI 318-14, Appendix A—Steel Reinforcement Information; ASTM standard reinforcing bars.



Plan View
(Bottom Reinforcement Layout)

Area inside the critical perimeter:

$$A_o = (c + d)^2$$

$$A_o = (39 + 26.5)^2$$

$$A_o = 4290.25 \text{ in}^2 = 29.79 \text{ ft}^2$$

Soil reaction inside:

$$P_{soil} = q_u A_o$$

$$P_{soil} = 2.303(29.793)$$

$$P_{soil} = 68.63 \text{ kip}$$

Therefore:

$$V_u = 419.8502 - 68.63$$

$$V_u = 351.22 \text{ kip}$$

For square interior pedestal:

Therefore:

$$\beta = \frac{39}{39} = 1.0$$

So $\beta = 1$means the loaded area is square (The design is limited to square cross-section).

$$\beta = 1$$

location of pedestal

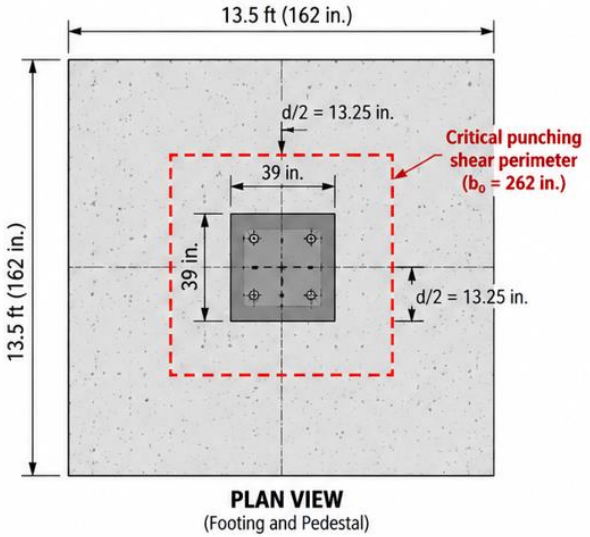
$\alpha_s$depends on where the loaded area is located (ACI 318-19 22.6.5.3):

Pedestal location	α_s
Interior	**40**
Edge	**30**
Corner	**20**

Your footing is being designed as an interior/square pedestal condition, so:

$\alpha_s = 40$

Equation 1: where $\beta \leq 2$



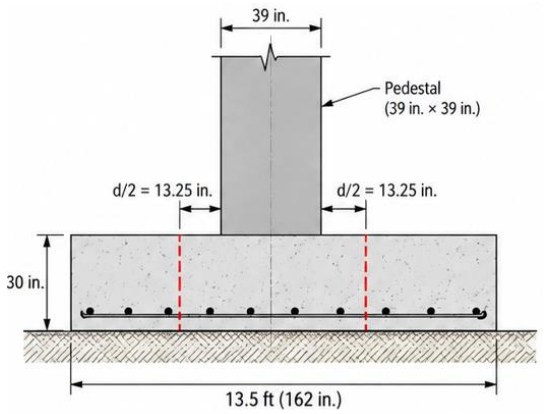


Table 19.2.4.1(a)—Values of λ for lightweight concrete based on equilibrium density

w_c, lb/ft³	λ	
≤ 100	0.75	(a)
100 < w_c ≤ 135	0.0075w_c ≤ 1.0	(b)
> 135	1.0	(c)

$$\phi V_c = 0.75\left(4\lambda_s\lambda\sqrt{f_c'}\right)b_o d$$

However, for **isolated footings**, the size-effect factor may be taken as:

$$\lambda = 1.0$$

The size-effect factor may be neglected for shallow foundations based ACI-318-19.

Thus:

$$\phi V_c = 0.75(4)(1)(1)\sqrt{4000}(262)(26.5)$$

$$\phi V_{c1} = 1317.3\ kip$$

The other two expressions give approximately:

Equation 3

$$\phi V_c = 0.75\left(2 + \frac{\alpha_s d}{b_o}\right)\lambda_s\lambda\sqrt{f'_c}\,b_o d$$

$$\phi V_{c3} = \frac{0.75\left(2 + \frac{40(26.5)}{262}\right)(1)(1)\sqrt{4000}(262)(26.5)}{1000}$$

$$\phi V_{c3} = 1993.0 kip$$

Therefore:

$$\phi V_c = 1317.3\ kip$$

Check:

$$351.22 < 1317.3 \ldots\ldots \textit{PUNCHING SHEAR — PASS}$$

Utilization:

$$\frac{351.22}{1317.3} = 0.2665$$

$$\textit{Utilization} = 26.65\%$$

Note: User Can Reduce footing width to increase utilization.

Step 7: One-Way Shear

The footing projection from the pedestal face to the footing edge is:

$$l = \frac{B - c}{2} = \frac{13.5 - 3.25}{2} = 5.125\ ft$$

The critical section for one-way shear is located at a distance d from the face of the pedestal.

$$d = h - c_c - \frac{d_b}{2}$$

$$d = 30 - 3 - \frac{1.0}{2} = 26.5\ in. = 2.208\ ft$$

The factored soil pressure obtained previously is:

$$q_u = 2.303\ ksf$$

Therefore, for a 1-ft-wide strip:

$$V_u = q_u(l - d)(1)$$

$$V_u = (2.303)(5.125 - 2.208)(1)$$

$$V_u = 6.718\ kip$$

For normal-weight concrete, $\lambda = 1.0$, and:

$$V_c = 2\lambda\sqrt{f_c'}b_w d$$

$$V_c = 2(1.0)\sqrt{4000}(12)(26.5)$$

$$V_c = 40.22\ kip$$

With $\phi = 0.75$:

$$\phi V_c = 0.75(40.22)$$

$$\phi V_c = 30.17\ kip$$

Check:

$$\phi V_c = 30.17 > V_u = 6.718\ kip \text{..... } \textit{One-way shear check PASS}$$

Utilization:

$$\frac{6.718}{30.17} = 0.223$$

$$\textbf{Utilization} = \mathbf{22.3}\%$$

Step 8: Flexure and Reinforcement

For a 12-in.-wide strip:

$$b = 12\ in.$$

$$d = 26.5\ in$$

$$f_c' = 4.0\ ksi$$

$$f_y = 60\ ksi$$

$$\phi = 0.90$$

For a uniformly distributed factored soil pressure:

$$M_u = \frac{ql^2}{2}$$

$$M_u = \frac{2.303(5.125)^2}{2}$$

$$M_u = \ 30.24\ kip - ft/ft$$

$$\boldsymbol{M_u = 362.88\ \text{kip} - \text{in./ft}}$$

the nominal moment strength is:

$$M_n = A_s f_y \left(d - \frac{a}{2}\right)$$

where

$$a = \frac{A_s f_y}{0.85 f_c' b}$$

Therefore:

$$\phi A_s f_y \left[d - \frac{A_s f_y}{2(0.85 f_c' b)}\right] = M_u$$

Substituting:

$$0.90 A_s (60) \left[26.5 - \frac{A_s (60)}{2(0.85)(4)(12)}\right] = 358.16$$

Solving for A_s:

$$A_{s,req} = \frac{0.252\ in^2}{ft}$$

Minimum reinforcement

$$A_{s,min} = \rho_{min} bh$$

Using:

$$\rho_{min} = 0.0018$$
$$A_{s,min} = (0.0018)(12)(30)$$
$$A_{s,min} = \frac{0.648\ in^2}{ft}$$

Therefore, the governing required reinforcement is:

$$A_{s,design} = \max(A_{s,req}, A_{s,min})$$
$$A_{s,design} = \max(0.252, 0.648)$$
$$A_{s,design} = \frac{0.648\ in^2}{ft}$$

Provided reinforcement

For a No. 8 bar:

$$A_b = 0.79\ in^2$$

Select:

$$\#8@14\ in.\ o.c.\ each\ way$$

The reinforcement provided per foot is:

$$A_{s,prov} = A_b \frac{12}{s}$$
$$A_{s,prov} = 0.79 \frac{12}{14}$$
$$A_{s,prov} = \frac{0.677\ in^2}{ft}$$

Check against required reinforcement:

$$0.677 > 0.648$$
$$A_{s,prov} \geq A_{s,design}\ PASS$$

Flexural strength check

For the provided reinforcement:

$$a = \frac{(0.677)(60)}{(0.85)(4)(12)}$$

$$a = 0.996 \text{ in.}$$

Therefore:

$$M_n = (0.677)(60)\left(26.5 - \frac{0.996}{2}\right)$$

$$M_n = 1056.4 \; kip\text{-}in./ft$$

and:

$$\phi M_n = 0.90(1056.4)$$

$$\phi M_n = 950.8 \; kip\text{-}in./ft$$

Compare:

$$\phi M_n = 950.8 > M_u = 358.2 \; kip\text{-}in./ft$$

$$\therefore \; \textbf{\textit{FLEXURE PASS}}$$

Final design

Bottom reinforcement = #8 @ 14 in. o.c. each way

$$\boldsymbol{A_{s,\text{prov}} = 0.\frac{677 \text{ in}^2}{\text{f}}}$$

$$\boldsymbol{\phi M_n = 950.8 \text{ kip-in./ft} > M_u = 358.2 \text{ kip-in./ft}}$$

FLEXURE: PASS

Step 9: Pedestal/Footing Bearing

$$N_1 = \phi(0.85 f_c' A_1)$$

with:

$$\phi = 0.65$$

For your 39 × 39 pedestal:

$$A_1 = 39(39) = 1521 \; in^2$$

Therefore:

$$N_1 = 0.65(0.85)(4)(1521)$$

$$N_1 = 3361.4\ kip$$

Compare:

$$P_u = 520\ kip$$

$$520 < 3361.4 \ldots\ldots\ldots \textit{PEDESTAL BEARING — PASS}$$

Step 10: Dowel / Pedestal-to-Footing Reinforcement

For a cast-in-place pedestal connected to a foundation, ACI 318-14 $16.3.4.1 states:

$$A_{s,\text{dowel}} \geq 0.005A_g$$

where A_g is the gross cross-sectional area of the supported pedestal/column and at least four dowels.

Based on pedestal:

$$A_{s,min} = 0.005(1521)$$

$$A_{s,min} = 7.605\ in^2$$

Adopted 18#6:

$$A_s = 18(0.44)$$

$$A_s = 7.92\ in^2$$

Therefore:

$$7.92 > 7.605 \ldots\ldots \textit{DOWEL/INTERFACE STEEL — PASS}$$

Step 11: Development of deformed bars in compression

$$l_{dc} = \frac{f_y \psi_r}{50\lambda\sqrt{f_c'}} d_b$$

with the minimum requirement:

$$0.0003 f_y d_b \geq 8\ in.$$

For #6:

$$d_b = 0.75\ in.$$

$$l_{dc} = \frac{60{,}000(1.0)}{50(1)\left(\sqrt{4000}\right)}(0.75)$$

$l_{dc} \approx 14.23$ in.

Minimum:

$$0.0003(60,000)(0.75) = 13.5\ in.$$

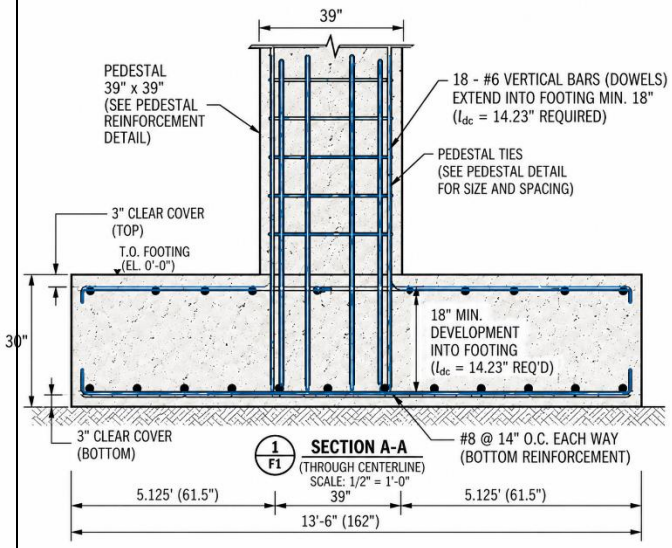
Therefore:

$$l_{dc} = 14.23\ in$$

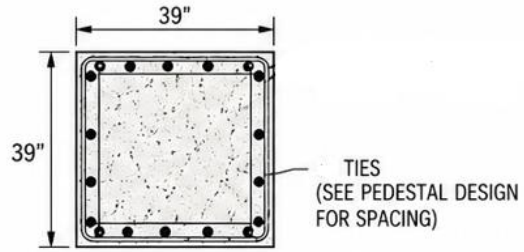
For detailing, we can adopt:

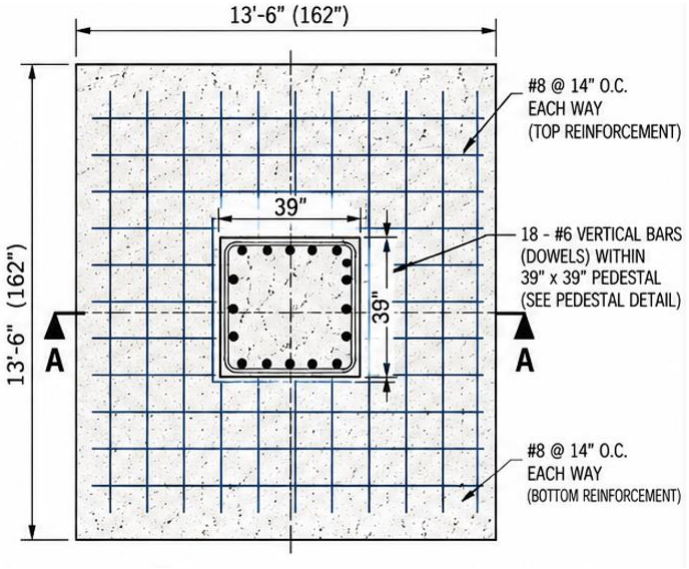
$$18 > 14.23$$

18 *in. minimum developed length*









BAR SIZE INFORMATION (ASTM A615)		
Bar No.	Diameter, d_b (in.)	Area, A_b (in.^{2})
#6	0.750	0.44
#8	1.000	0.79